

Root [0]

Technical Sheet Only!
Please refer to the subsequent sheets for schematic

FPGA



File: fpga.kicad_sch

Connectivity



File: Connectivity.kicad_sch

Reviewer: Michal Varsanyi
Designer: Michal Varsanyi
Czech Technical University in Prague

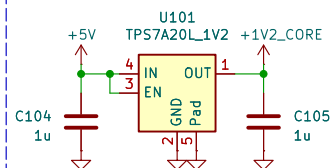
Sheet: /
File: Core_Module.kicad_sch

Title: Root

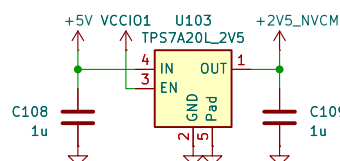
Size: A4	Date: 2026-03-31	Rev: 2.0
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FPGA [1]

Core Supply Voltage



NVCM Supply

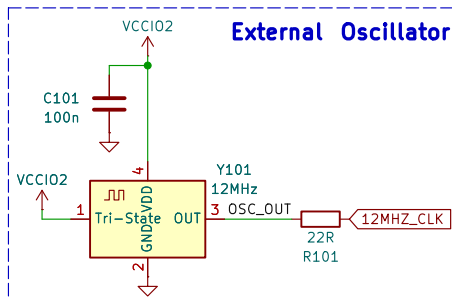


The NVCM regulator can be omitted and the VPP2V5 pin can be connected directly to 3V3 in order to save cost.

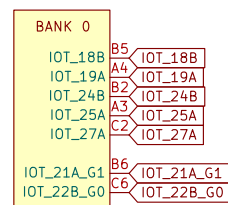
The pin can handle 3.6V.

A diode can be used to drop the voltage before by roughly 0.7V

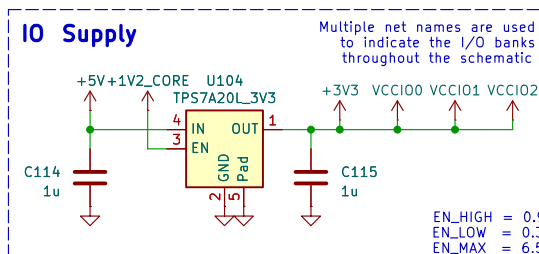
External Oscillator



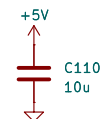
Bank 0 I/O



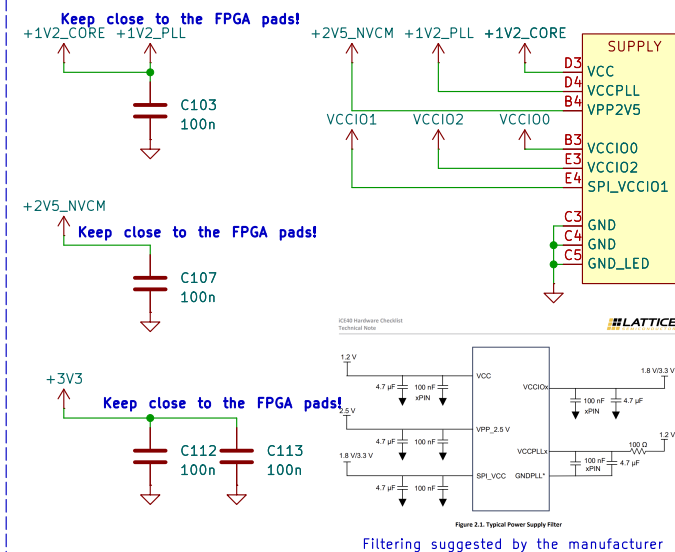
10 Supply



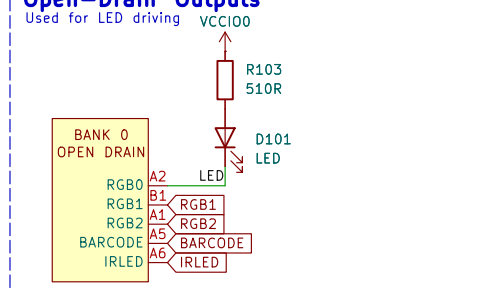
Input Snubber



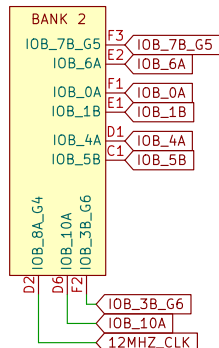
FPGA Power Input



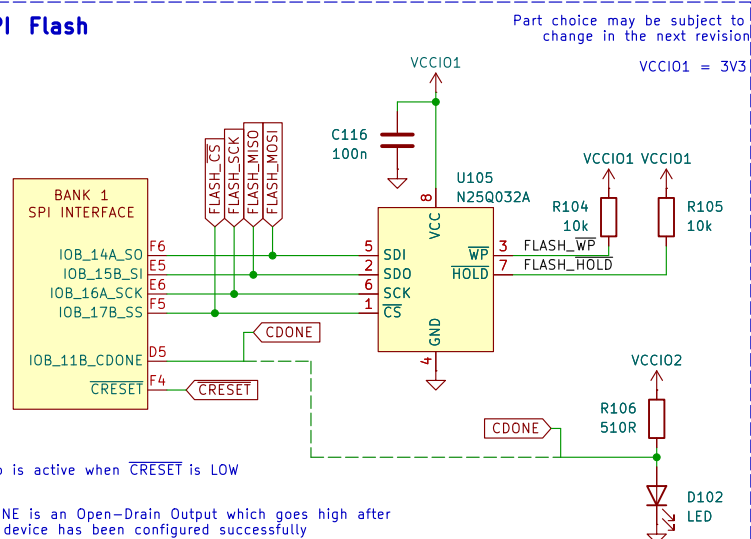
Open-Drain Outputs



Bank 2 I/O



SPI Flash



Chip is active when $\overline{\text{CRESET}}$ is LOW

CDONE is an Open-Drain Output which goes high after the device has been configured successfully

See: iCE40 Programming and Configuration app note
Section 9.2. SPI PROM Requirements

If the LED shines, the device has been configured properly

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Title: FPGA

Size: A4	Date: 2026-03-31
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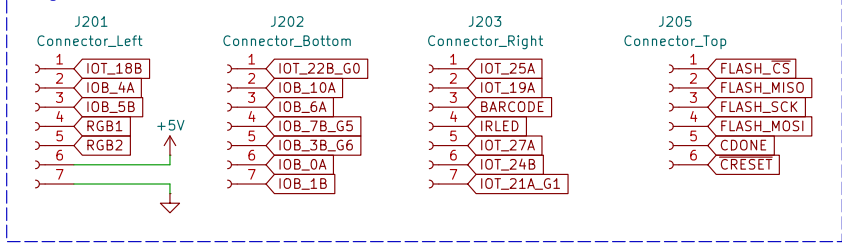
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Rev: 2.0

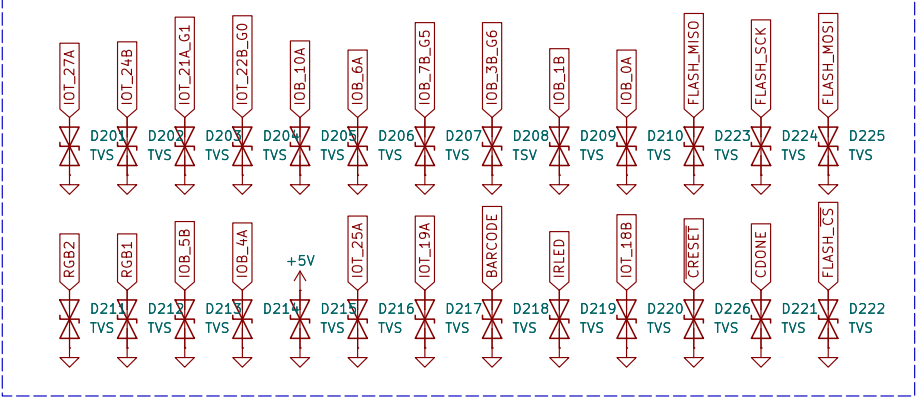
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Connectivity [2]

Edge Connectors



ESD Protection



Reviewer: Michal Varsanyi
Designer: Michal Varsanyi
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Sheet: /Connectivity/
File: Connectivity.kicad_sch

Title: Connectivity

Size: A4 Date: 2026-03-31
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Rev: 2.0
Id: 2/3